

updated: 7/25

Curriculum Vitae: Chad McKell

ABOUT

Position	Research Engineer
Company	Sigma Connectivity, on assignment at Meta
Address	10301 Willows Road NE, Redmond, WA 98052
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Research	My research spans mathematical modeling and numerical simulation of physical systems with an emphasis on acoustics and audio. Recent projects include geometric boundary modeling for acoustic wave simulation; non-spherical harmonics for sound capture; and elastic wave modeling for sound synthesis and auditory biophysical analysis. Past projects include underwater acoustic sensing and optical standing-wave trapping.
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EDUCATION

9/19–	University of California San Diego , Ph.D. in Computer Music Specialized in Computational Acoustics and Applied Mathematics Dissertation: <i>Conformal Symmetry for Exterior Wave Problems</i> . Advisors: Albert Chern (Computer Science) and Miller Puckette (Music).
9/16–10/17	University of Edinburgh , M.S. in Acoustics and Music Technology
8/09–12/15	Wake Forest University , M.S. in Physics
6/02–8/09	Brigham Young University , B.S. in Biophysics

EMPLOYMENT

9/24–	Sigma Connectivity, on assignment at Meta , Research Engineer
3/24–6/24	University of California San Diego , Teaching Assistant in Computer Science
6/23–2/24	Meta, Reality Labs Research , Research Scientist Intern
8/21–3/22	Meta, Reality Labs Research , Research Intern
9/19–6/23	University of California San Diego , Teaching Assistant/Researcher in Music
7/18–7/19	Applied Research in Acoustics , R&D Scientist
5/18–5/18	Moog Music (self-employed) , Audio Software Developer
4/17–9/17	Lofelt, acquired by Meta in 2022 (self-employed) , Acoustics Researcher
10/14–8/16	J.P. Morgan/Neovest (via ConsultNet) , Software Development Engineer in Test
8/12–12/12	University of North Carolina School of the Arts , Adjunct Instructor of Physics
9/09–9/11	Wake Forest University , Teaching Assistant in Physics
9/08–6/09	Brigham Young University , Tutorial Lab Assistant in Physics
8/07–3/09	Brigham Young University , Research Assistant in Philosophy

RESEARCH ACTIVITIES

9/24–	Sigma Connectivity, on assignment at Meta , Research Engineer Redmond, Washington. Affiliation: Research Audio. Research topics: <i>spatial audio</i> , <i>software engineering</i> . Research summary: Build novel spatial audio technology for Reality Labs Research at Meta. Supervisor: Sebastian Prepelitã. Team Lead: Ravish Mehra. I work onsite at Meta as a vendor employed by Sigma Connectivity.
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RESEARCH ACTIVITIES CONT.

- 9/19–9/24 **University of California San Diego**, Ph.D. Student
La Jolla, California. Affiliations: Department of Music and Center for Visual Computing. Research topics: *computational physics, numerical simulation, differential geometry*. Research summary: developed mathematical and computational methods for solving exterior wave problems. Committee members: Albert Chern (co-chair), Miller Puckette (co-chair), Melvin Leok, Shahrokh Yadegari, Stefan Bilbao (Univ. of Edinburgh), and Sebastian Prepelitã (Meta).
- 6/23–2/24 **Meta, Reality Labs Research**, Research Scientist Intern
Redmond, Washington. Affiliation: Research Audio. Research topics: *spatial audio, software engineering*. Research summary: Built novel spatial audio technology for applications in virtual and augmented reality. Supervisor: Sebastian Prepelitã. Team Lead: Ravish Mehra.
- 8/21–3/22 **Meta, Reality Labs Research**, Research Intern
La Jolla, California. Affiliation: Research Audio. Research description: see above.
- 7/18–7/19 **Applied Research in Acoustics**, R&D Scientist
Culpeper, Virginia. Research topics: *underwater acoustic sensing, acoustic beamforming*. Research summary: developed physics-based signal processing algorithms for object detection and classification in naval sonar systems. Team Lead: Jonathan Botts.
- 1/17–8/17 **University of Edinburgh**, Master’s Student
Edinburgh, Scotland. Affiliation: Reid School of Music. Research topics: *computational acoustics, elastodynamics, numerical simulation*. Research summary: developed numerical simulations of structural acoustic vibrations for commercial haptic devices. My thesis was partially funded by Lofelt, a Berlin-based haptic feedback company acquired by Meta in 2022. Advisor: Stefan Bilbao.
- 1/10–9/13 **Wake Forest University**, Master’s Student
Winston-Salem, North Carolina. Affiliation: Department of Physics. Research topics: *optical trapping, laser beam characterization, fluid dynamics*. Research summary: simulated and fabricated standing-wave optical traps for Brownian particle tracking. Advisor: Keith Bonin.
- 8/07–8/09 **Brigham Young University**, Undergraduate Student
Provo, Utah. Affiliation: Department of Physiology and Developmental Biology. Research topics: *biophysics, scanning probe microscopy*. Research summary: investigated structural effects of isoflurane on artificial lipid bilayers using atomic force microscopy. Advisor: David Busath.

TEACHING EXPERIENCE

As Instructor

UNCSA
SCI 1100

General Physics. Fall 2012 (1 term).

TEACHING EXPERIENCE CONT.

As TA

UCSD

CSE 291	Physics Simulation. Spring 2024 (1 term).
MUS 5	Sound in Time. Spring 2020 (1 term).
MUS 6	Electronic Music. Fall 2020 (1 term).
MUS 15	Popular Music: David Bowie. Winter 2021 (1 term).
MUS 15	Popular Music: Video Game Music. Winter 2020 (1 term).
MUS 171	Computer Music I. Winter 2022 (1 term).
MUS 172	Computer Music II. 2021–2022 (2 terms).

WFU

PHY 113	General Physics I (Mechanics). 2009–2011 (4 terms).
PHY 114	General Physics II (E&M). Fall 2010 (1 term).

As Tutor

BYU

PHSCS 105	General Physics 1 (Mechanics). 2008–2009 (2 terms).
PHSCS 106	General Physics 2 (E&M). Winter 2009 (1 term).
PHSCS 121	Principles of Physics 1 (Mechanics). 2008–2009 (2 terms).
PHSCS 123	Principles of Physics 2 (Waves/Thermo). W/Sp 2009 (2 terms).
PHSCS 220	Principles of Physics 3 (E&M). W/Sp 2009 (2 terms).

PUBLICATIONS

Manuscripts in Progress

- (1) **C. McKell**, M. Nabizadeh, S. Wang, and A. Chern, “Wave simulations in infinite spacetime”.

Simulating wave propagation on an infinite domain has been a long-standing computational challenge. Conventional approaches to this problem only produce wave simulations on a small subset of the infinite domain. Using the fact that wave propagation on an infinite Minkowski spacetime is equivalent to wave propagation on a bounded Minkowski spacetime under a Kelvin-like transformation, we simulate wave propagation on the entire infinite domain using a finite discretization of the bounded domain with no additional loss of accuracy from the transformation.

- (2) **C. McKell**, *Conformal Symmetry for Exterior Wave Problems*, Ph.D. Dissertation, University of California San Diego. Defense planned for Fall 2025. Advisors: Albert Chern and Miller Puckette.

This work presents novel geometric methods for handling open exterior boundaries in wave simulations. First, I discuss extensions to the reflectionless discrete perfectly matched layer for the wave and Helmholtz equations. Then, I demonstrate that the conformal invariance of the wave equation under a Kelvin transform in Minkowski spacetime allows one to convert an infinite domain problem into a bounded domain problem that can be solved using standard numerical methods with no additional loss of accuracy introduced by the transform. I conclude by discussing parallel computing strategies for the geometric boundary models, applications in architectural acoustics and binaural audio, and future work in obstacle boundary flattening.

PUBLICATIONS CONT.

Journal Articles

- (3) **C. McKell** and K. Bonin, “Optical corral using a standing-wave Bessel beam,” *Journal of the Optical Society of America B*, Vol. 35, No. 8, 1910–1920, 2018.

Conference Proceedings

- (4) **C. McKell**, “Sonification of optically-ordered Brownian motion,” In Proceedings of the International Computer Music Conference (ICMC), Utrecht, Netherlands, September 2016.

Master’s Theses

- (5) **C. McKell**, *Real-Time Physical Modeling for Haptic Feedback Rendering*, Master’s Thesis, University of Edinburgh, Acoustics and Audio Group, 2017. Advisor: Stefan Bilbao.
- (6) **C. McKell**, *Confinement and Tracking of Brownian Particles in a Bessel Beam Standing Wave*, Master’s Thesis, Wake Forest University, Department of Physics, 2015. Advisor: Keith Bonin.

Technical Reports

- (7) **C. McKell**, H. Conley, and D. Busath, “AFM study of structural changes in supported planar DPPC bilayers containing general anesthetic isoflurane,” Brigham Young University, Paper 827, 2010.